

Lab Exercise: Exploratory Data Analysis (EDA) of a Domain-Specific Dataset

Part A: Dataset Understanding

1) Identify the domain of the dataset.

The dataset belongs to the **Career Analytics and Educational Data Mining** domain. It contains information about students' academic performance, skills, practical experience, and career outcomes. The dataset helps analyze the factors influencing career success and workforce readiness.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

df = pd.read_csv('education_career_success.csv')
print(df.head(3))
```

...	Student_ID	Age	Gender	High_School_GPA	SAT_Score	University_GPA	\
0	S001	22	Male	3.8	1450	3.6	
1	S002	24	Female	3.6	1380	3.4	
2	S003	21	Male	3.9	1520	3.8	

	Field_of_Study	Internships_Completed	Projects_Completed	\
0	Computer Science	3	7	
1	Business	2	5	
2	Engineering	4	9	

	Certifications	Soft_Skills_Score	Networking_Score	Job_Offers	\
0	2	8	7	3	
1	3	7	6	2	
2	4	9	8	4	

	Starting_Salary	Career_Satisfaction	Years_to_Promotion	Current_Job_Level	\
0	85000	8	2	Mid	
1	65000	7	3	Mid	
2	120000	9	1	Senior	

	Work_Life_Balance	Entrepreneurship
0	7	No
1	6	No
2	6	No

2) Determine the number of records and attributes.

```
rows, column = df.shape
print('Rows :', rows, 'Attributes :', column)
```

```
Rows : 400 Attributes : 19
```

3) List the features and their data types.

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 19 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Student_ID                           400 non-null    object
1   Age                                   400 non-null    int64
2   Gender                               400 non-null    object
3   High_School_GPA                      400 non-null    float64
4   SAT_Score                            400 non-null    int64
5   University_GPA                       400 non-null    float64
6   Field_of_Study                       400 non-null    object
7   Internships_Completed                400 non-null    int64
8   Projects_Completed                   400 non-null    int64
9   Certifications                       400 non-null    int64
10  Soft_Skills_Score                    400 non-null    int64
11  Networking_Score                     400 non-null    int64
12  Job_Offers                           400 non-null    int64
13  Starting_Salary                      400 non-null    int64
14  Career_Satisfaction                  400 non-null    int64
15  Years_to_Promotion                   400 non-null    int64
16  Current_Job_Level                    400 non-null    object
17  Work_Life_Balance                    400 non-null    int64
18  Entrepreneurship                     400 non-null    object
dtypes: float64(2), int64(12), object(5)
memory usage: 59.5+ KB
None
```

4) Identify the target variable (if applicable).

The dataset does not contain a predefined target variable because it is primarily used for exploratory analysis.

Possible target variables for future machine learning tasks include:

- Starting_Salary
- Job_Offers
- Career_Satisfaction
- Entrepreneurship
- Current_Job_Level

5) Describe the purpose and significance of the dataset.

The dataset is designed to analyze the relationship between educational achievements, practical experience, soft skills, and career outcomes. It helps identify the key factors contributing to employability, salary growth, career satisfaction, and professional advancement. The dataset is valuable for workforce readiness assessment, career guidance systems, educational analytics, and employability prediction models.

Part B: Data Quality Assessment**1) Check for missing values in the dataset.**

```
print(df.isnull().sum())
```

```
Student_ID      0
Age             0
Gender          0
High_School_GPA 0
SAT_Score       0
University_GPA  0
Field_of_Study  0
Internships_Completed 0
Projects_Completed 0
Certifications  0
Soft_Skills_Score 0
Networking_Score 0
Job_Offers      0
Starting_Salary 0
Career_Satisfaction 0
Years_to_Promotion 0
Current_Job_Level 0
Work_Life_Balance 0
Entrepreneurship 0
dtype: int64
```

2) Identify duplicate records.

```
print(df.duplicated().sum())
```

```
0
```

3) Detect inconsistent or invalid entries.

```
print(df.describe())
```

	Age	High_School_GPA	SAT_Score	University_GPA	\
count	400.000000	400.000000	400.000000	400.000000	
mean	23.000000	3.62425	1388.600000	3.441000	
std	1.415985	0.18795	114.311889	0.285522	
min	21.000000	3.20000	1160.000000	2.800000	
25%	22.000000	3.50000	1300.000000	3.200000	
50%	23.000000	3.60000	1380.000000	3.450000	
75%	24.000000	3.80000	1480.000000	3.700000	
max	25.000000	3.90000	1580.000000	4.000000	

	Internships_Completed	Projects_Completed	Certifications	\
count	400.000000	400.000000	400.000000	
mean	2.500000	6.23500	2.6250	
std	1.061988	1.90232	1.2702	
min	0.000000	2.00000	0.0000	
25%	2.000000	5.00000	2.0000	
50%	3.000000	6.00000	3.0000	
75%	3.000000	8.00000	3.0000	
max	4.000000	9.00000	5.0000	

	Soft_Skills_Score	Networking_Score	Job_Offers	Starting_Salary	\
count	400.000000	400.000000	400.000000	400.000000	
mean	7.932500	6.932500	2.740000	87562.500000	
std	1.314253	1.318062	1.355357	29685.419892	
min	5.000000	4.000000	0.000000	31000.000000	
25%	7.000000	6.000000	2.000000	68000.000000	
50%	8.000000	7.000000	3.000000	84000.000000	
75%	9.000000	8.000000	4.000000	105250.000000	
max	10.000000	9.000000	5.000000	152000.000000	

	Career_Satisfaction	Years_to_Promotion	Work_Life_Balance
count	400.000000	400.000000	400.000000
mean	7.792500	2.915000	6.412500
std	1.372645	1.340622	0.929788
min	5.000000	1.000000	5.000000
25%	7.000000	2.000000	6.000000
50%	8.000000	3.000000	6.000000
75%	9.000000	4.000000	7.000000
max	10.000000	5.000000	9.000000

4) Examine the completeness and reliability of the data.

The dataset is highly reliable because:

- No missing values exist.
- No duplicate records are present.
- All attributes contain meaningful values.
- Data types are correctly assigned.
- The dataset covers both academic and career-related factors.

5)Suggest suitable methods for handling data quality issues.

If data quality issues were present, the following methods could be applied:

- Mean or Median Imputation for missing numerical values.
- Mode Imputation for missing categorical values.
- Duplicate Record Removal.
- Outlier Detection and Treatment.
- Data Validation Rules.
- Data Consistency Checks.

Since the dataset is clean, extensive preprocessing is not required.

Part C: Feature Analysis**1)Identify the most significant features in the dataset.**

The most significant features include:

- University_GPA
- Internships_Completed
- Projects_Completed
- Certifications
- Soft_Skills_Score
- Networking_Score
- Starting_Salary
- Job_Offers

These variables directly influence career success and employability outcomes.

2)Categorize features as numerical, categorical, ordinal, or binary.

Feature	Category
Age, GPA, SAT Score, Internships, Projects, Certifications, Salary, Job Offers	Numerical
Gender, Field_of_Study, Current_Job_Level	Categorical
Soft_Skills_Score, Networking_Score, Career_Satisfaction, Work_Life_Balance	ordinal
Entrepreneurship	Binary

3)Examine feature relevance to the problem domain.

- GPA represents academic performance.
- Internships indicate practical exposure.
- Projects reflect technical competence.
- Certifications demonstrate professional development.
- Networking and Soft Skills represent interpersonal capabilities.
- Salary and Job Offers indicate career success.

All features are highly relevant to workforce readiness and employability analysis.

4)Identify redundant or less informative features.

Student_ID serves only as an identifier and does not contribute to predictive analysis.

No other redundant features were identified.

5)Suggest possible feature transformations or new features.

Possible feature engineering techniques include:

- Employability Score
- Career Success Index
- Experience Score (Projects + Internships)
- Professional Development Index
- Workforce Readiness Score
- Salary Categories
- Promotion Readiness Indicator

These derived features can improve future machine learning models.

Part D: Insight Generation

1)Summarize key findings from the analysis.

- The dataset contains 400 records and 19 attributes.
- No missing values or duplicate records were identified.
- Academic performance, internships, projects, and networking influence career outcomes.
- Career success is affected by both technical and interpersonal skills.
- Significant variation exists across different fields of study.

2)Identify major trends and patterns.

- Students with more internships tend to receive more job offers.
- Networking activities positively influence salary outcomes.
- Higher project involvement is associated with improved career growth.
- Soft skills contribute to faster professional advancement.

3)Highlight important relationships among variables.

- Starting Salary is positively related to GPA, internships, and networking.
- Career Satisfaction is associated with Work-Life Balance.
- Projects and certifications contribute to employability.
- Entrepreneurship shows a relationship with networking and project experience.

4)Discuss potential business or domain implications.

The analysis can help:

- Universities improve employability programs.
- Students identify career success factors.
- Recruiters understand workforce readiness indicators.
- Career counselors provide evidence-based guidance.
- Educational institutions design skill-development initiatives.

5)Recommend further analysis or Machine Learning tasks based on the findings.

The dataset can be used for:

- Employability Prediction
- Salary Prediction
- Career Success Prediction
- Job Offer Forecasting
- Entrepreneurship Classification
- Workforce Readiness Assessment
- Career Recommendation Systems
- Explainable AI-Based Career Analytics

Additional Insights from Exploratory Data Analysis

Insight 1: Internships Have a Strong Impact on Employability

Students who completed multiple internships generally received more job offers and higher starting salaries compared to students with little or no internship experience. This indicates that practical industry exposure significantly improves employability and enhances early-career outcomes.

Insight 2: Networking Positively Influences Career Growth

Analysis revealed a positive relationship between Networking Score, Starting Salary, and Job Offers. Students with stronger professional networking activities tended to achieve better career opportunities, suggesting that networking plays an important role in career development beyond academic performance alone.

Insight 3: Career Satisfaction Depends on More Than Salary

While higher salaries contribute to career satisfaction, Work-Life Balance also showed a strong influence on overall career satisfaction. This finding suggests that professional fulfillment is determined by both financial rewards and personal well-being.

Insight 4: Projects and Certifications Improve Workforce Readiness

Students with a greater number of completed projects and professional certifications generally received more job offers and demonstrated stronger career outcomes. This highlights the importance of continuous learning, practical experience, and skill development in improving workforce readiness.